

भारतीय मानक
Indian Standard

IS 2112 : 2014

(Reaffirmed 2019)

चाँदी एवं चाँदी मिश्रधातुएँ
आभूषण/शिल्पकारी — शुद्धता
एवं मुहरांकन — विशिष्टि
(तीसरा पुनरीक्षण)

**Silver and Silver Alloys, Jewellery/
Artefacts — Fineness and Mark-
ing — Specification**
(Third Revision)

ICS 39.060

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Price Group 1

Precious Metals Sectional Committee, MTD 10

FOREWORD

This Indian Standard (Third Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Precious Metals Sectional Committee had been approved by the Metallurgical Engineering Division Council.

This standard was first published in 1962, and later revised in 1981 and 2003 respectively. The fineness of silver and silver alloys specified in this standard are in line with the fineness of silver and silver alloys in ISO 9202 : 1991 'Jewellery — Fineness of precious metals alloys'.

In many countries, the silver jewellery/artefacts (coins, medals, utensils, etc) trade are certified for purity and are marked with various details like purity/fineness of the article, manufacturer's mark, the mark of the assaying laboratory responsible for certifying the purity, etc. These jewellery/artefacts are termed as Hallmarked.

The need for availability of jewellery/artefacts of silver of certified purity in the country, for domestic use and for export is the major reason for undertaking the preparation of this standard.

This standard is primarily meant to classify the various grades of silver according to their silver content and to provide guidelines.

In this revision following changes have been made:

- a) All amendments issued earlier have been incorporated;
- b) A new grade of fineness 990 has been introduced;
- c) Clauses **1.2** 'Note', **5**, **6.3** and **6.3.1** have been modified; and
- d) Clauses **7.1** and **7.3** have also been modified.

For the purpose of deciding whether particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1060 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

**SILVER AND SILVER ALLOYS,
JEWELLERY/ARTEFACTS — FINENESS AND
MARKING — SPECIFICATION**

(Third Revision)

1 SCOPE

1.1 This standard specifies:

- a) Three grades of fine silver, and
- b) Six grades of silver alloys used in the manufacture of jewellery/artefacts of silver based on their silver content.

1.2 This standard also specifies the guidelines for marking of purity and other details on tested silver jewellery/artefacts.

NOTE — Artefacts cover medallions, utensils and coins provided the purity of silver alloy for artefacts is as per 5.

2 REFERENCES

The following standards listed below contain provisions which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards listed below:

<i>IS No.</i>	<i>Title</i>
1387 : 1993	General requirements for the supply of metallurgical materials (<i>second revision</i>)
2113 : 2002	Assaying silver in silver and silver alloys — Methods (<i>third revision</i>)

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Assaying — The method of accurate determination of silver content of the sample expressed in parts per thousand (‰).

3.2 Fineness — The ratio between the mass of silver content and total mass expressed in parts per thousand (‰).

3.3 Silver — The metallic element silver, free from any other element.

3.4 Silver Alloy — Silver alloyed with one or more other elements (*see* Note in 5).

4 SUPPLY OF MATERIAL

General requirements relating to the supply of material are laid down in IS 1387.

5 GRADES

Fine silver and silver alloys shall be classified in accordance with the following grades depending upon their fineness:

<i>Grade</i>		<i>Fineness</i>
9 999	Fine silver	999.9
9 995		999.5
999		999.0
990	Silver alloys for jewellery artefacts	990.0
970		970.0
925		925.0
900		900.0
835		835.0
800		800.0

NOTE — The above classification of silver alloys is applicable for silver jewellery/artefacts also. All the above grades of silver and silver alloys shall be free from cadmium and lead.

6 REQUIREMENTS

6.1 Silver alloys used for manufacture of jewellery/artefacts shall have the fineness as given in 5 without any negative tolerance.

6.1.1 Solder

In the manufacture of jewellery/artefacts of grades 900, 925, 970 and 990, the solder of purity 750 silver minimum and for grades of 800 and 835, the solder of purity 650 silver minimum free from cadmium and lead shall be used. However, melting purity of articles shall not be less than the purity for which it has been offered for hallmarking as per the grades given in 5.

6.2 Sampling

Unless otherwise specified for the jewellery/artefacts of the same design and quality, the sample shall be taken from 10 percent of the semi-finished/finished jewellery/artefacts. Any other sampling plan may also be followed subject to the agreement between the

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contracting parties or as specified by the inspection agency as the case may be.

6.3 Assaying

The sample taken as prescribed in 6.2 shall be assayed in accordance with IS 2113. The actual weight of samples, etc, shall however, be as prescribed in respective test methods as detailed in IS 2113.

6.3.1 Detection of cadmium and lead in silver and silver alloys shall be carried out by X-ray fluorescence (XRF) method till an established test procedure is formulated.

6.4 Tolerance

No negative tolerance shall be allowed on the silver content of silver jewellery/artefact sampled and tested as per 6.2 and 6.3 respectively.

7 MARKING

7.1 The silver and silver alloys, jewellery/artefacts shall be stamped with the Standard Mark in the case known as the 'HALLMARK' by BIS recognized Assaying and Hallmarking Centre only. The Hallmarked articles shall carry the following markings:

- a) BIS Mark along with the word 'SILVER';
- b) Purity grade/fineness;
- c) Assay centre's identification mark;
- d) Year of marking denoted by a letter symbol (as defined by BIS); and
- e) Jeweller's mark/manufacturer's identification mark.

7.2 Hallmarking

The use of Hallmark is governed by the provisions of the *Bureau of Indian Standards Act*, 1986 and Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Hallmark may be granted to jewellers or jewellery/artefact manufacturers may be obtained from the Bureau of Indian Standards.

7.2.1 Only linear designs shall be permitted.

7.2.2 Markings shall be legible and durable. Marking shall be visible by a magnifying glass of magnification 10 X.

7.2.3 Markings shall be done on all parts which can be easily removed or replaced.

7.2.4 Silver jewellery/artefacts on which marking should not be applied are as follows:

- a) Any silver jewellery/artefact which has not been assayed by BIS recognized Assaying and Hallmarking Centre and does not confirm to the fineness/grade declared;
- b) Any silver jewellery/artefact which fails to comply with specified fineness;
- c) Any silver jewellery/artefact which is not solid or which is made with hollow centre and then filled with base metal, cement, lac or other substance thereby giving the article a false or added weight or strength; and
- d) Silver jewellery/artefact on which it is physically not possible to stamp the marks.

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Amendments Issued Since Publication

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